

Microprocessor & Computer Architecture

BCA — Semester II — 2021 — Answer Sheet

Subject Code: CACS155

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. Explain the addressing modes of 8085 with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Draw the timing diagram for MVI A, 35H and explain in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What do you mean by instruction mapping? Explain how to convert an instruction code to a microinstruction address.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define stack. Explain the stack organization.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain the arithmetic pipeline with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain the following instructions a) LDA 7050H b) CPI 35H c) PUSH B

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What are the typical characteristics of CISC instruction set architecture? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define the term mnemonics, instruction cycle and machine cycle. There are two tables holding twenty data whose starting address is 3000H and 3020H respectively. Write a program to add the content of first table with the content of second table having same array index. Store sum and carry into the third and fourth table indexing from 3040H and 3060H respectively.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. For the expression $Y = (A + B) * (C + D)$ Write down the code for one address, two address and three address instruction formats.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is parallel processing? Explain the benefits of parallel processing. Explain the classifications of parallel processing by M. J. Flynn.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.